

Stylometry

Simone Rebora

(Mendenhall, 1887)

THE IDEA

«Measuring» authorial style

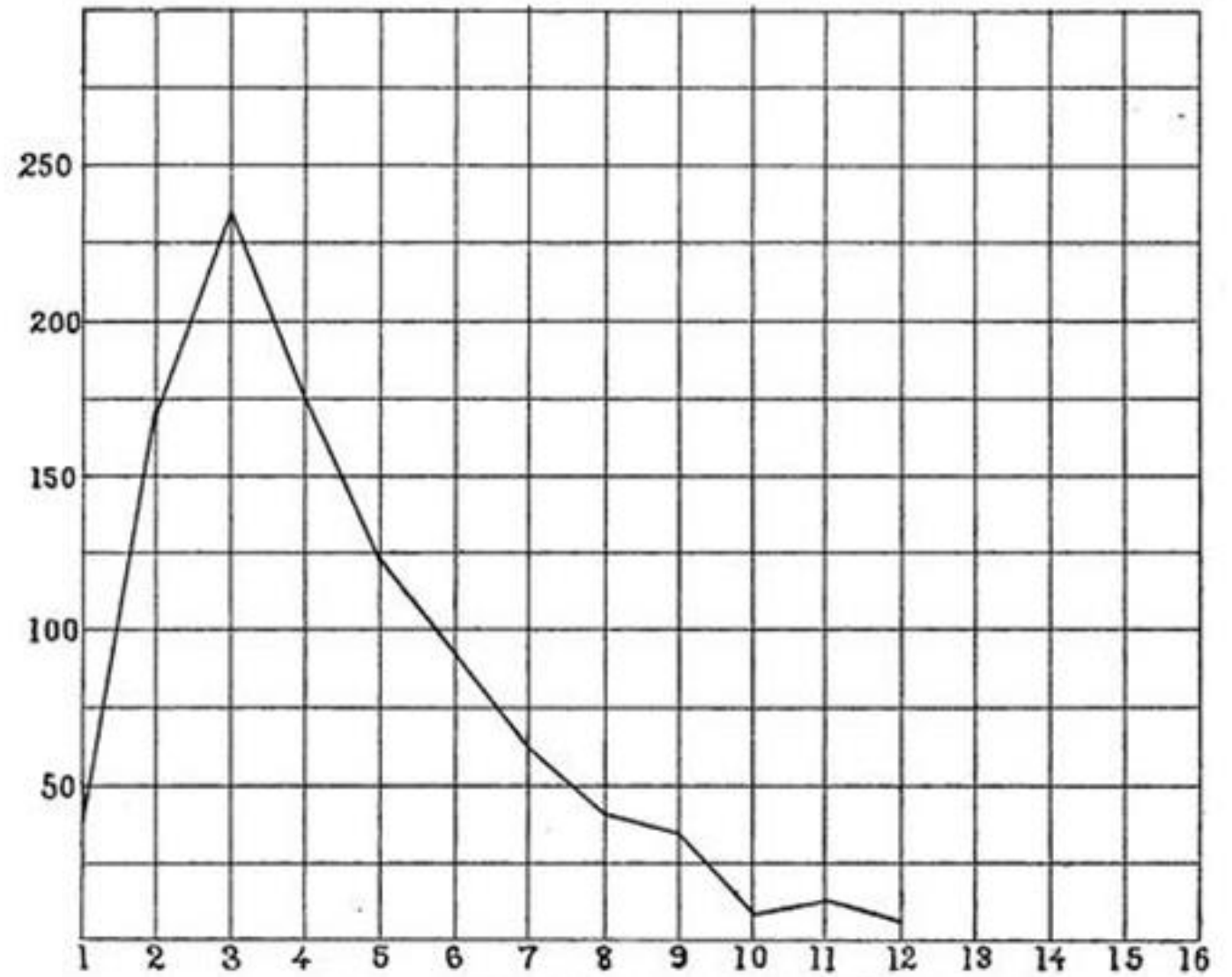
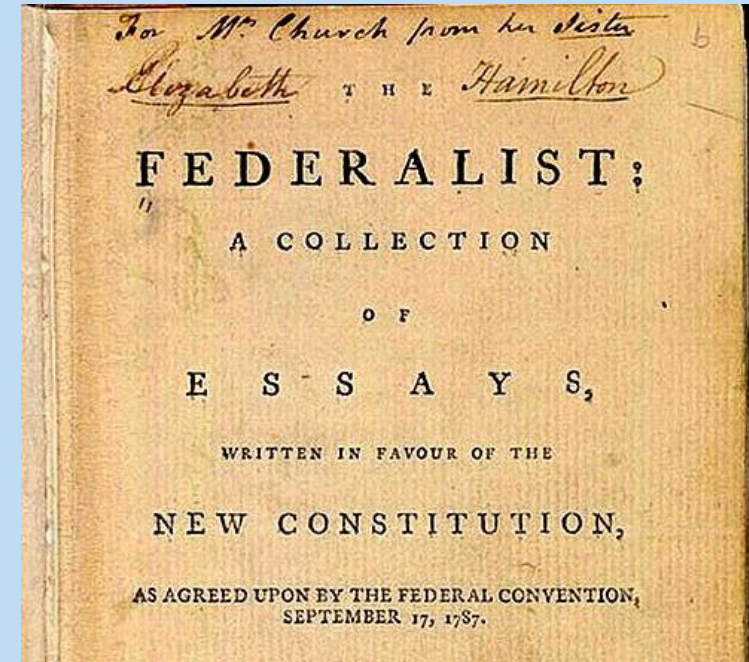


FIG. 1. — FIRST ONE THOUSAND WORDS IN 'OLIVER TWIST.'

THE IDEA

«Measuring» authorial style



...example of their procedure:

	enough	while	whilst	upon
Hamilton	0.59	0.26	0	2.93
Madison	0	0	0.47	0.16
Disputed texts	0	0	0.34	0.08

THE IDEA

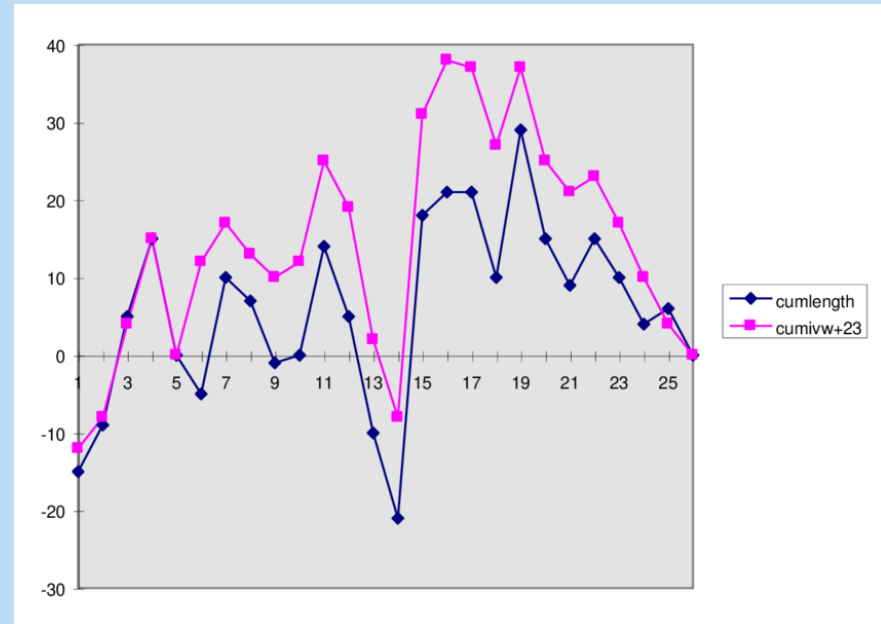
«Measuring» authorial style

Failures

Andrew Morton in the early '60 adapted **Cumulative Sum** – CUSUM or QSUM

During a BBC live show (1993):

Documents of convicted criminals were attributed to ... the Secretary of State for Justice!!! (cf. Holmes, 1998)



(Farrington et al. 1996)

THE (PLETHORA OF) METHODS FOR STYLOMETRY AND AUTHORSHIP ATTRIBUTION

- Character-level analysis
- Syntax-level analysis
- Multi-method analysis (e.g. JGAAP, PAN competition software...)
- ...and many others
- In this lesson, **just one method:**
 - Delta

WORD-FREQUENCY BASED STYLOMETRY

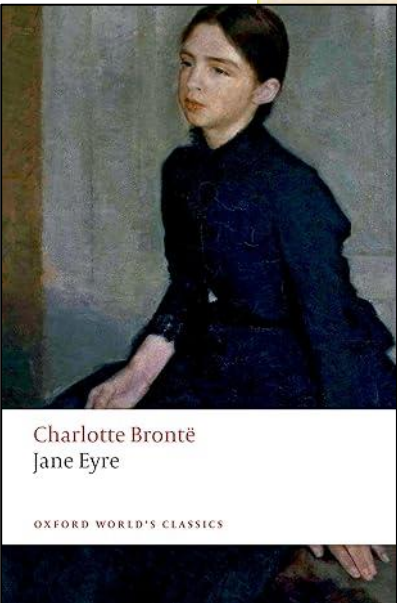
'Delta': a Measure of Stylistic Difference and a Guide to Likely Authorship¹

John Burrows
University of Newcastle, Australia

"Literary and
Linguistic
Computing"
17, no. 3
(2002): 267–
87

Abstract

This paper is a companion to my 'Questions of authorship: attribution and beyond', in which I sketched a new way of using the relative frequencies of the very common words for comparing written texts and testing their likely authorship. The main emphasis of that paper was not on the new procedure but on the broader consequences of our increasing sophistication in making such comparisons and the increasing (although never absolute) reliability of our inferences about authorship. My present objects, accordingly, are to give a more complete account of the procedure itself; to report the outcome of an extensive set of trials; and to consider the strengths and limitations of the new procedure. The procedure offers a simple but comparatively accurate addition to our current methods of distinguishing the most likely author of texts exceeding about 1,500 words in length. It is of even greater value as a method of reducing the field of likely candidates for texts of as little as 100 words in length. Not unexpectedly, it



1. the
2. and
3. of
4. to
5. a
6. i
7. in
8. he
9. was
10. it
11. that
12. you
13. his
14. her
15. with
16. as
17. had
18. she
19. for

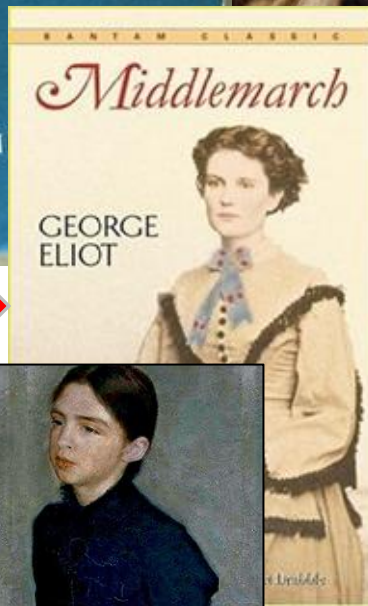
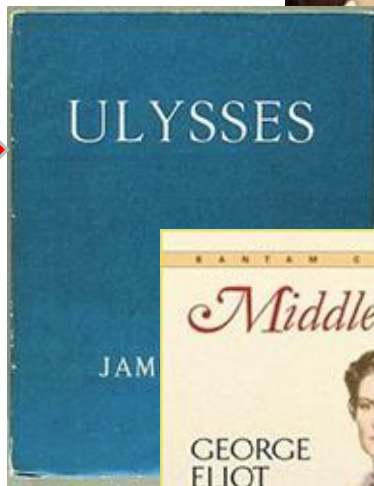
1. the
2. and
3. of
4. to
5. a
6. i
7. in
8. he
9. was
10. it
11. that
12. you
13. his
14. her
15. with
16. as
17. had
18. she
19. for

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2.4%
2.5%

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3.3%
2.2%
2.7%

3.1%
4.2%
1.4%
1.2%

5.2%
3.2%
2.4%
2.5%



$$\Delta_{(AB)} = \frac{1}{n} \sum_{i=1}^n \left| \frac{A_i - \mu_i}{\sigma_i} - \frac{B_i - \mu_i}{\sigma_i} \right|$$

	A	B	C	D	E	F
1		AlessandroManzoni_Adelchi	AlessandroManzoni_IlContediCarmagnola	AlessandroManzoni_InniSacri	AlessandroManzoni_Odi	AlessandroManzoni_Poesiegio
2	AlessandroManzoni_Adelchi	0	0,481290655	0,666926925	0,738545533	0,5688
3	AlessandroManzoni_IlContediCarmagnola	0,481290655	0	0,746348745	0,814261157	0,6545
4	AlessandroManzoni_InniSacri	0,666926925	0,746348745	0	0,633663965	0,6348
5	AlessandroManzoni_Odi	0,738545533	0,814261157	0,633663965	0	0,7338
6	AlessandroManzoni_Poesiegiovani	0,568820863	0,654375023	0,634854567	0,733827682	
7	CarloGoldoni_Gl'Innamorati	0,980786338	0,936018177	1,013723738	1,101305203	0,9504
8	CarloGoldoni_IlCampiello	1,016924762	1,031300757	1,018625104	1,092680684	0,9295
9	CarloGoldoni_II Servitore di due Padroni	0,94860233	0,926662976	0,976288639	1,080804722	0,9185
10	CarloGoldoni_IlTeatrocomico	0,915941412	0,896367382	0,971870697	1,085346366	0,8984
11	CarloGoldoni_IIVentaglio	1,011953514	1,00041649	1,074888328	1,131792245	0,9972
12	CarloGoldoni_IRusteghi	1,089096895	1,124315967	1,047451935	1,1240649	0,9778
13	CarloGoldoni_LaBottegadelcaff�	0,997940632	0,980781404	1,069965126	1,139058754	0,9938
14	CarloGoldoni_LaFamigliadell'Antiquario	0,97647637	0,968110166	1,038499373	1,080510085	0,9530
15	CarloGoldoni_LaLocandiera	0,97946604	0,952399004	1,052505983	1,110322738	0,9561
16	CarloGoldoni_LeBaruffechiozzotte	1,051753673	1,103993387	1,018834132	1,082447143	0,9425
17	CarloGoldoni_LeFemminepuntigliose	0,940334542	0,938723973	1,008461186	1,076438004	0,9179
18	CarloGoldoni_LeSmanieperlaVilleggiatura	1,023938091	0,964832878	1,056736183	1,148650567	1,0072
19	CarloGoldoni_UnadelleultimeserediCarnovale	1,045847956	1,085480986	1,047945641	1,10681856	0,9488
20	VittorioAlfieri_Agamennone	0,684514153	0,743793265	0,829452563	0,905939302	0,7061
21	VittorioAlfieri_Antigone	0,73781244	0,801189414	0,824156384	0,91495815	0,7219
22	VittorioAlfieri_Brutosecondo	0,675393312	0,675937144	0,830722082	0,910174086	0,6685
23	VittorioAlfieri_Filippo	0,69672213	0,73856813	0,806194725	0,93419818	0,6694
24	VittorioAlfieri_MariaStuarda	0,693145931	0,715015202	0,806081448	0,948928306	0,6738
25	VittorioAlfieri_Merope	0,735463235	0,783055974	0,855979157	0,971583955	0,7097
26	VittorioAlfieri_Mirra	0,76329317	0,819104452	0,864045202	0,9659327	0,7605
27	VittorioAlfieri_Oreste	0,70530237	0,777981376	0,829335057	0,930970217	0,7154
28	VittorioAlfieri_Ottavia	0,762895099	0,791949819	0,874379901	0,96265065	0,7225
29	VittorioAlfieri_Saul	0,645417404	0,735038238	0,760393582	0,871007648	0,6668
30						

	Berlin	Brussels	Dublin	Lyon	Madrid	Munich	Paris	Rome
Berlin	0	123	345	456	678	899	987	122
Brussels	123	0	363	255	151	515	515	156
Dublin	345	363	0	455	266	157	662	351
Lyon	456	255	455	0	683	156	167	642
Madrid	678	151	266	683	0	462	215	155
Munich	899	515	157	156	462	0	433	122
Paris	987	515	662	167	215	433	0	521
Rome	122	156	351	642	155	122	521	0

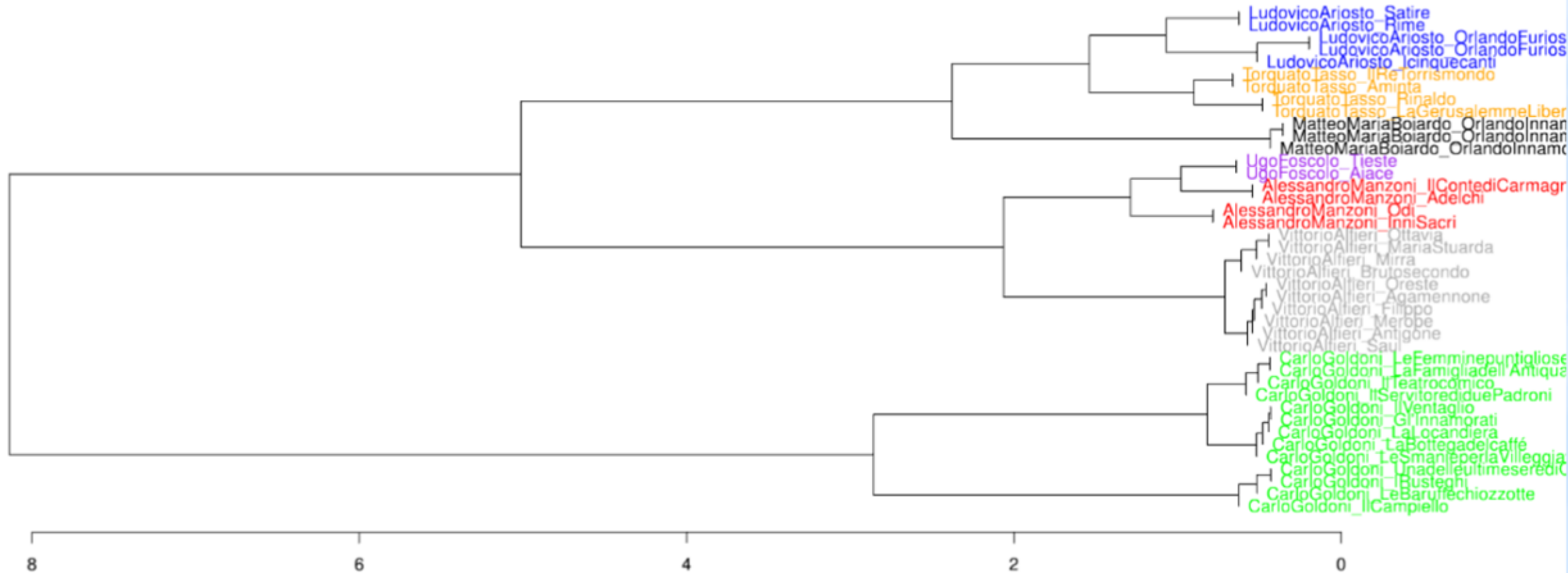


VISUALIZATIONS

1. Dendrograms

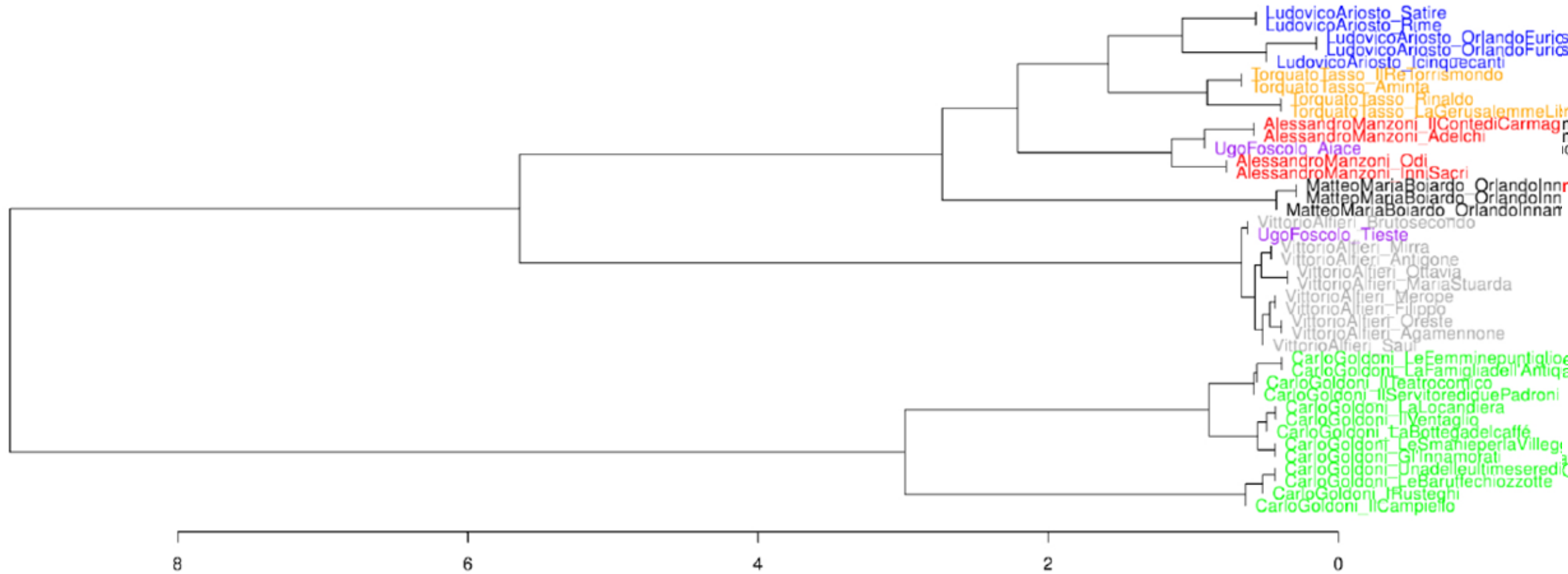
Ward's clustering algorithm (Ward, 1963)

Letteratura Italiana
Cluster Analysis



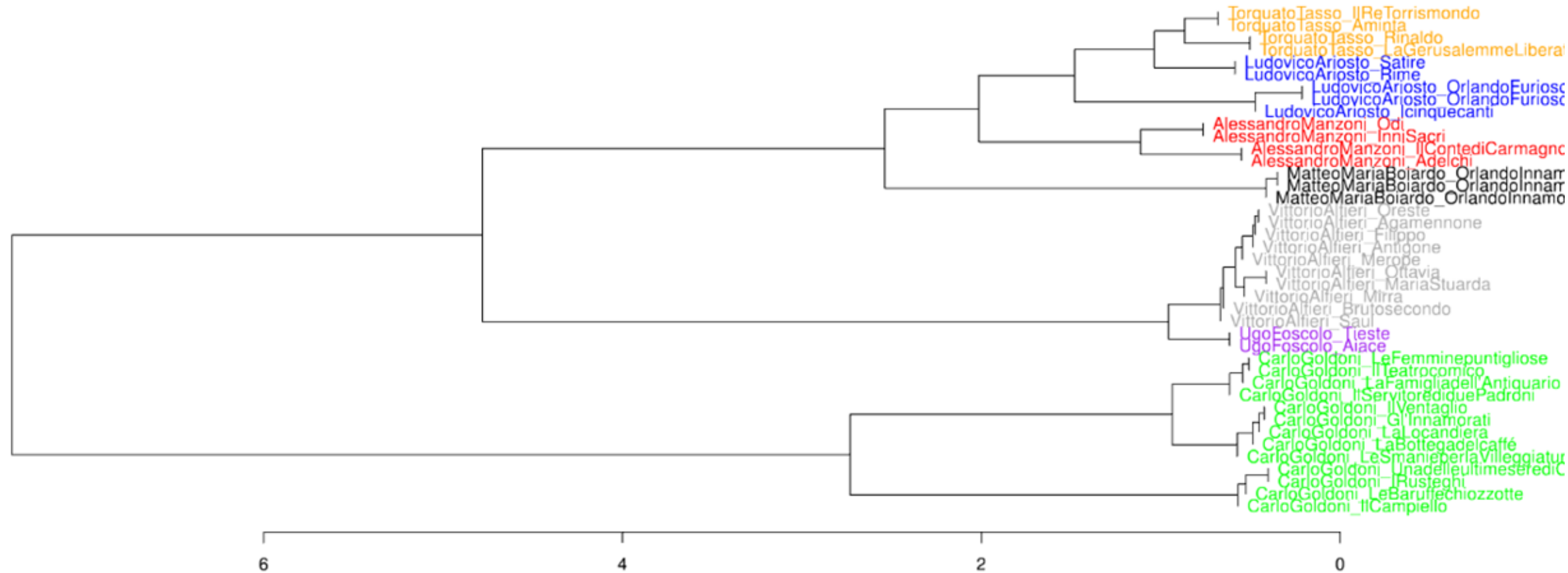
Burrows Delta
with 100 most frequent words (MFW)

Letteratura Italiana Cluster Analysis



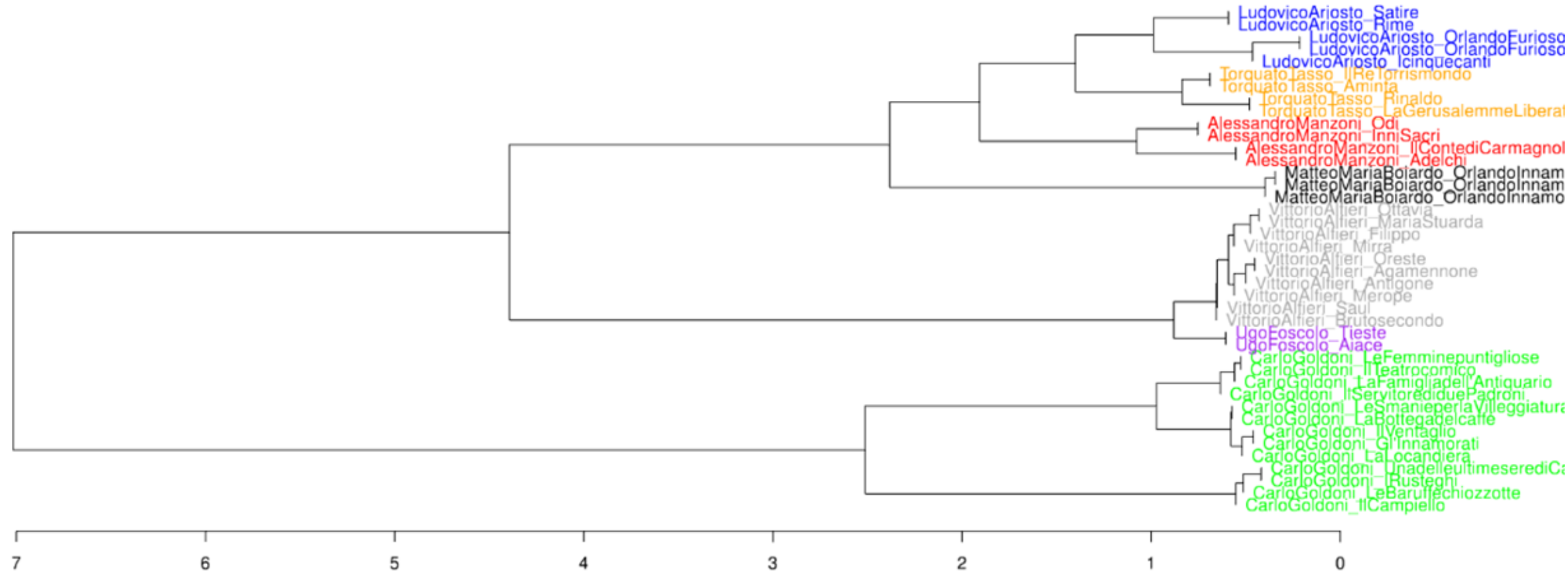
Burrows Delta
with 200 most frequent words (MFW)

Letteratura Italiana Cluster Analysis



Cosine Delta
with 100 most frequent words (MFW)

Letteratura Italiana
Cluster Analysis



My Weird Distance Measure
with 1,000,000 most frequent words (MFW)

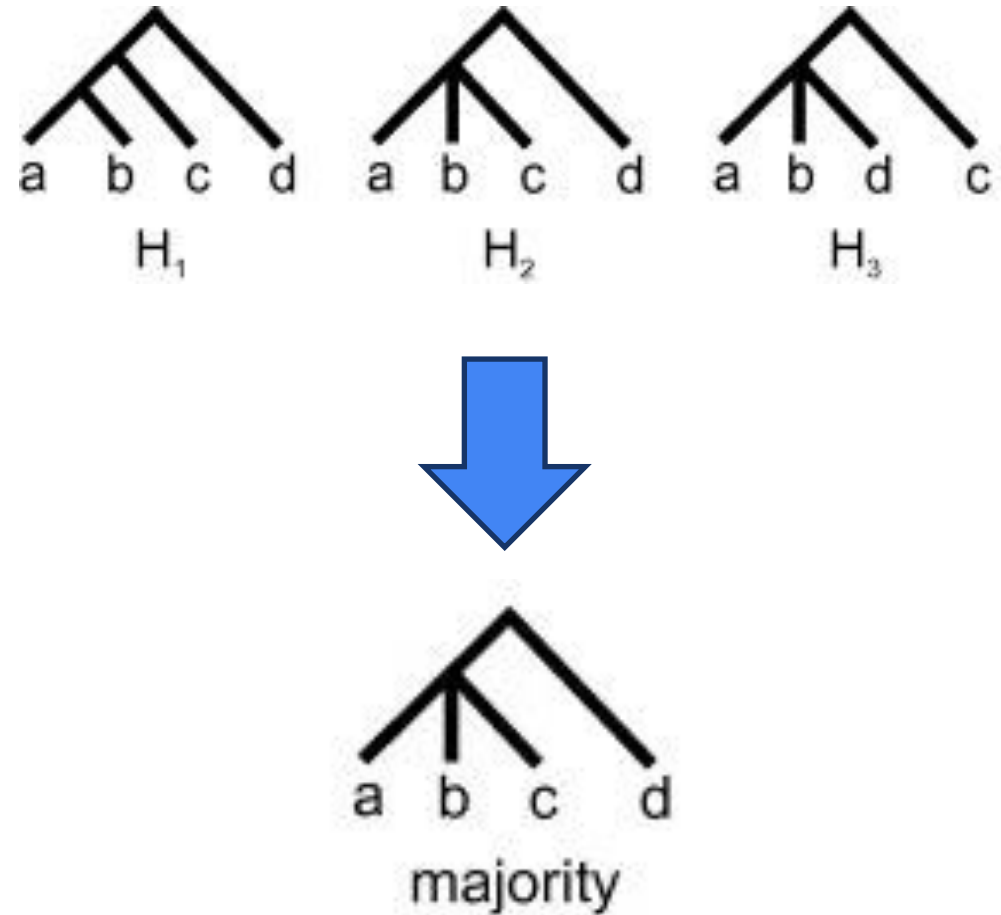
VISUALIZATIONS

2. Consensus Trees

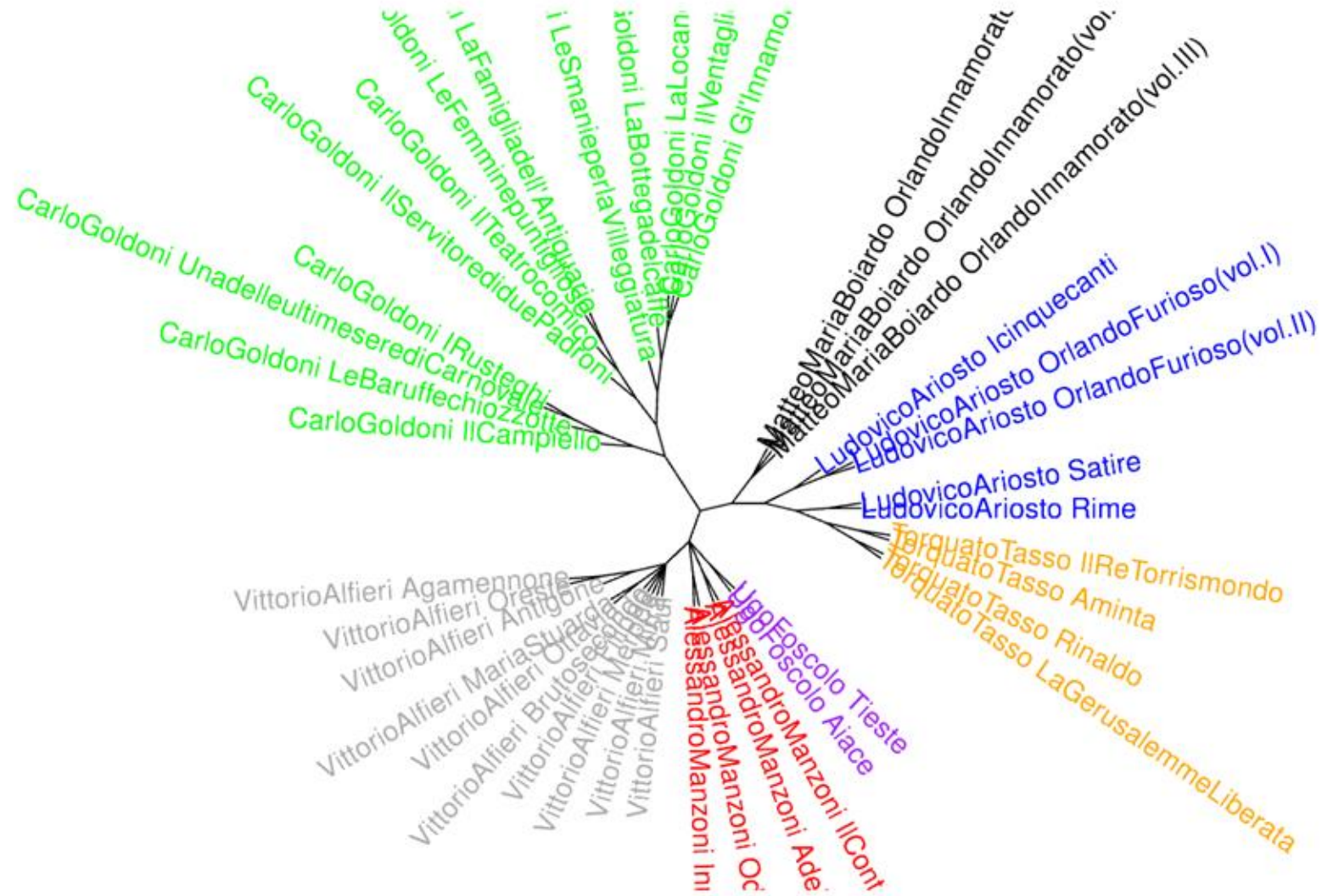
Method developed in phylogenetics
(see Paradis et al. 2004)

VISUALIZATIONS

Majority-rule consensus

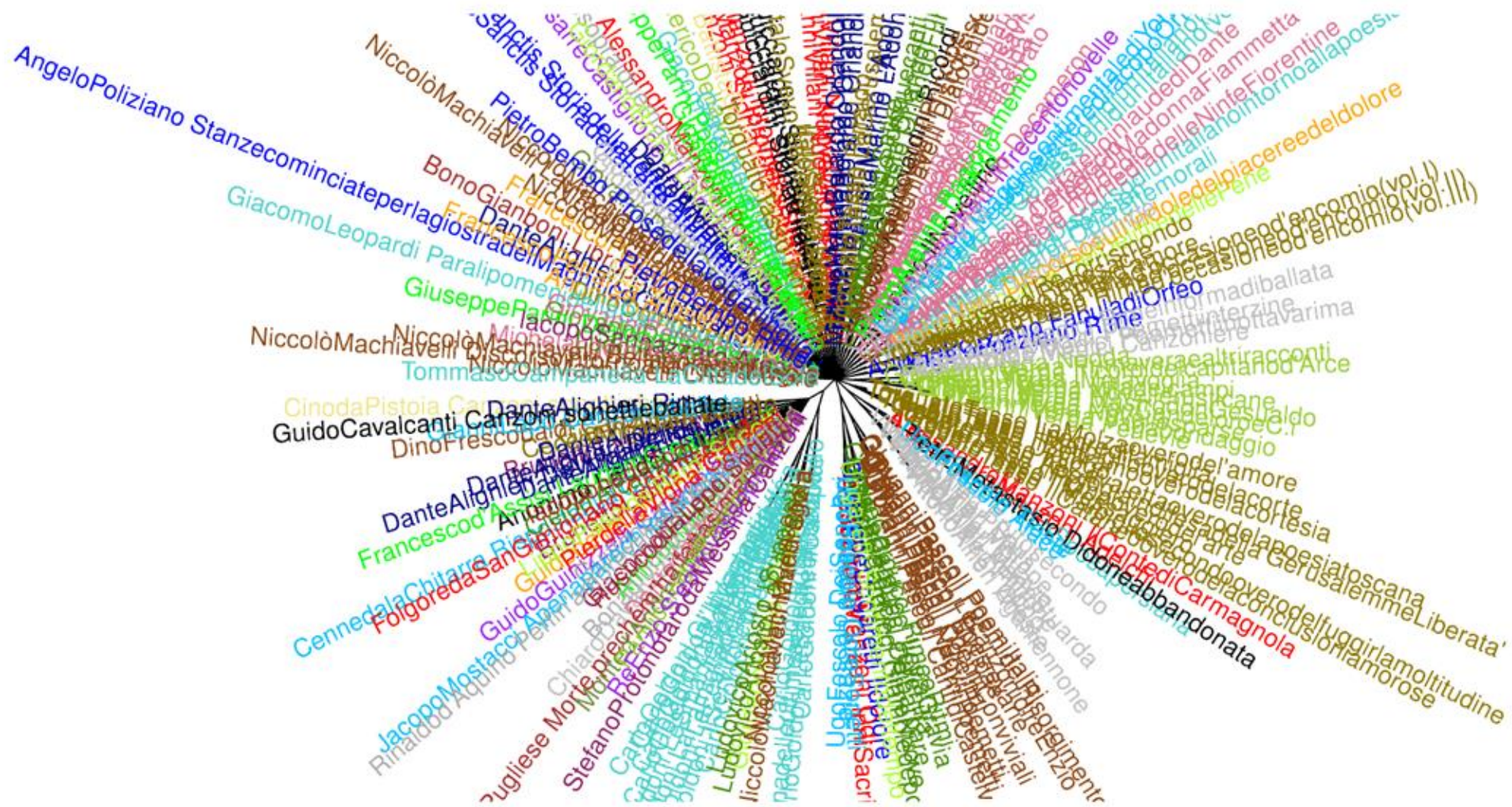


Letteratura Italiana Bootstrap Consensus Tree



100-1000 MFW Culled @ 0%
Classic Delta distance Consensus 0.5

Letteratura Italiana Bootstrap Consensus Tree



100-1000 MFW Culled @ 0%
Classic Delta distance Consensus 0.5

VISUALIZATIONS

3. Network Graphs

See Eder, 2017

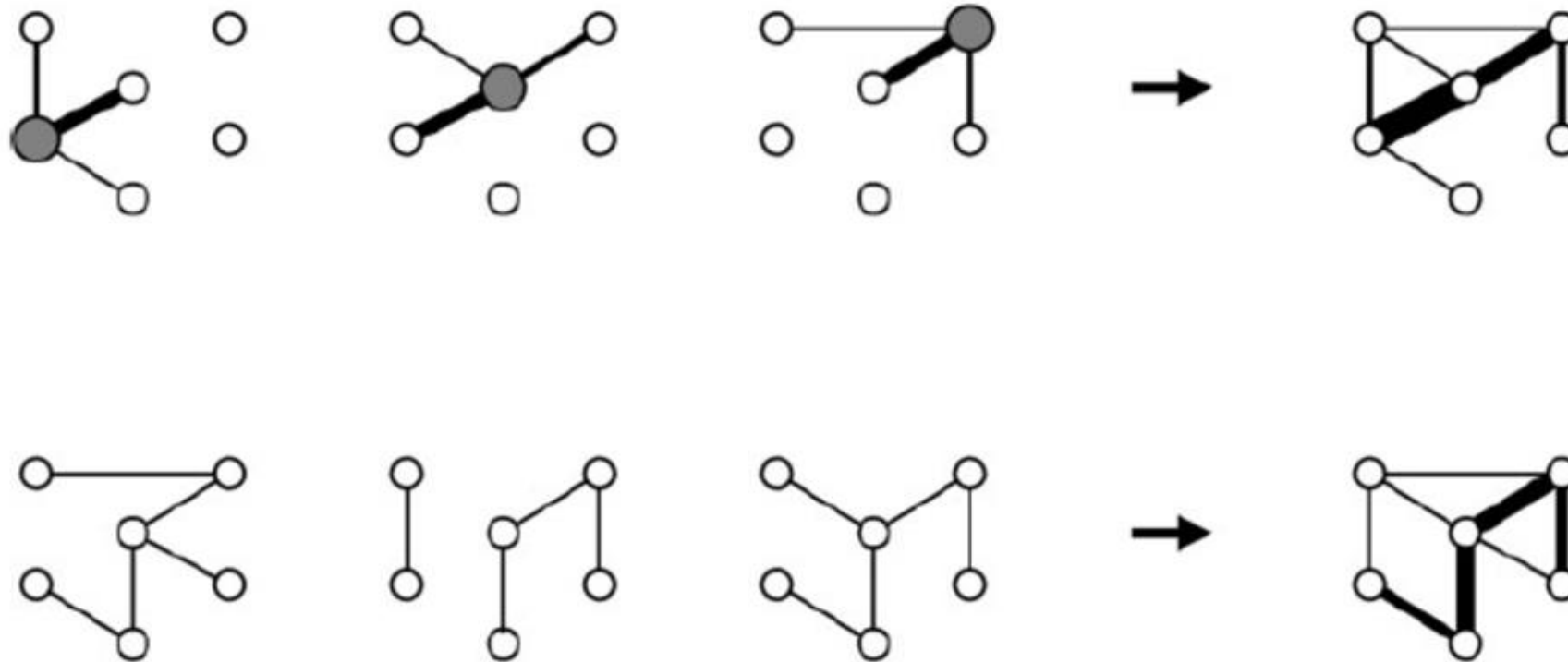
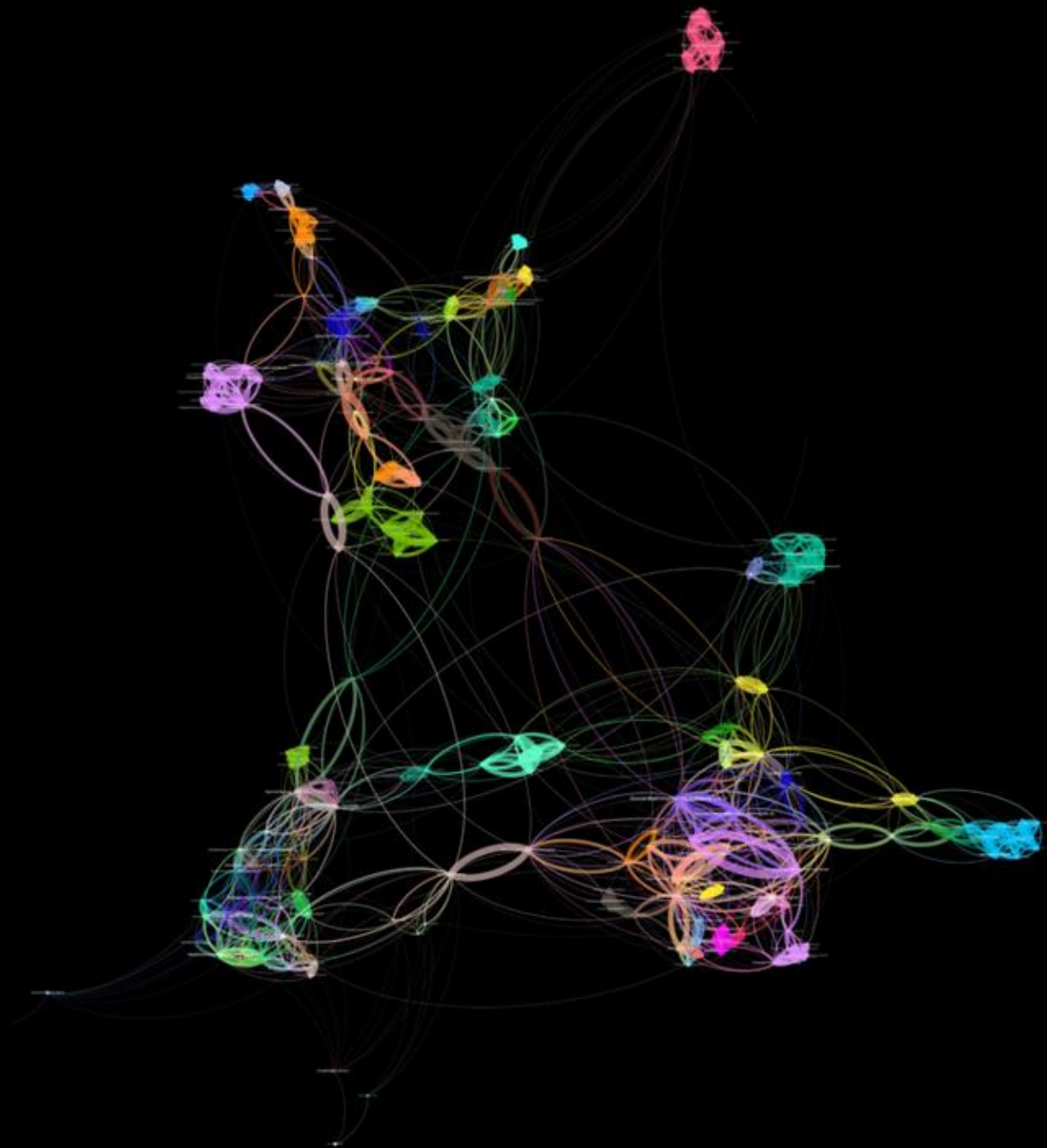
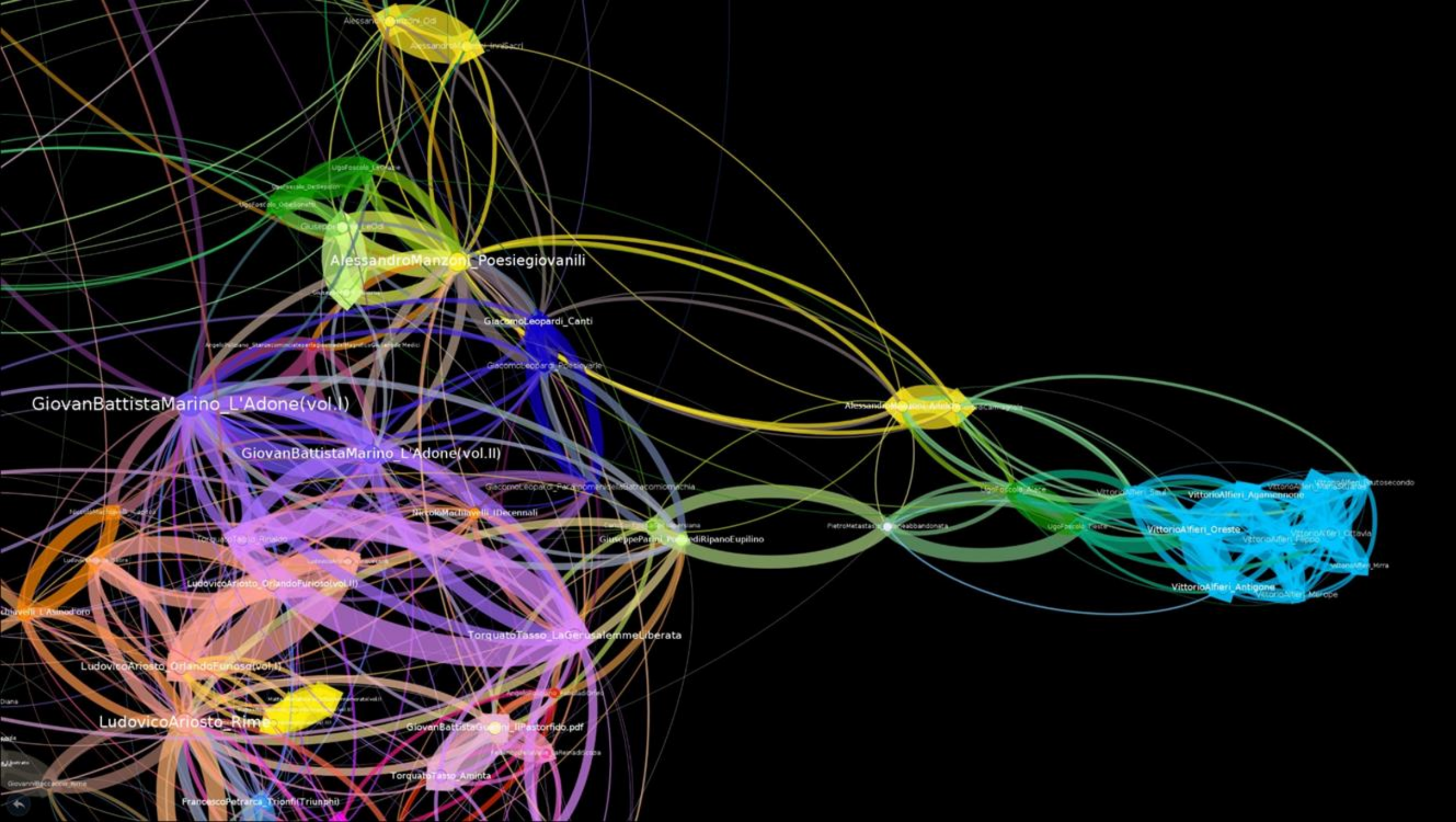
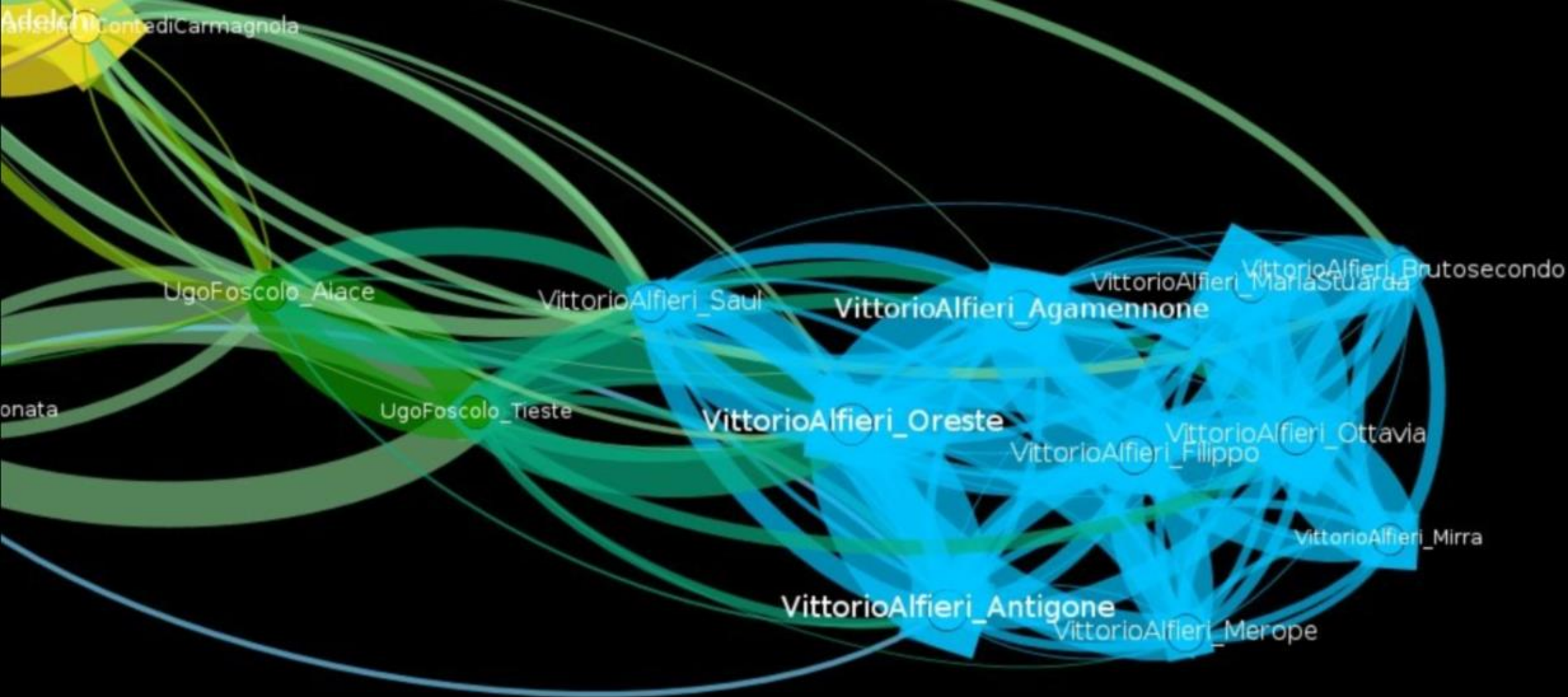


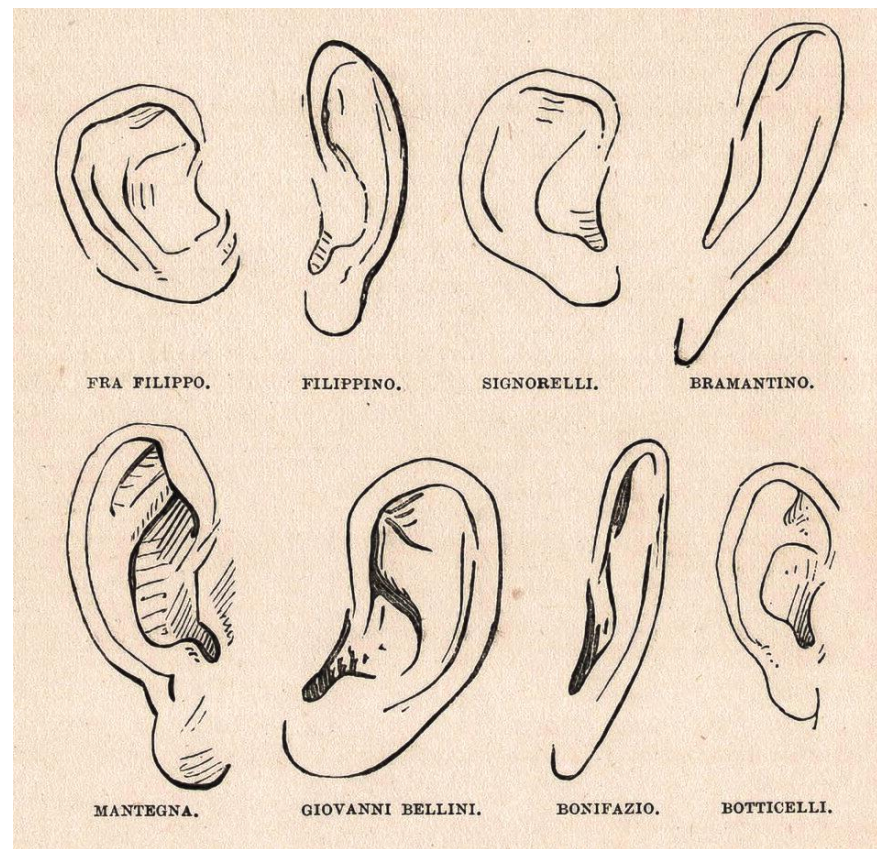
Fig. 6. Two algorithms of mapping textual relations: establishing weighted links to a nearest neighbor and two runners-up (top); producing a consensus network (bottom).







WHY DOES IT WORK?

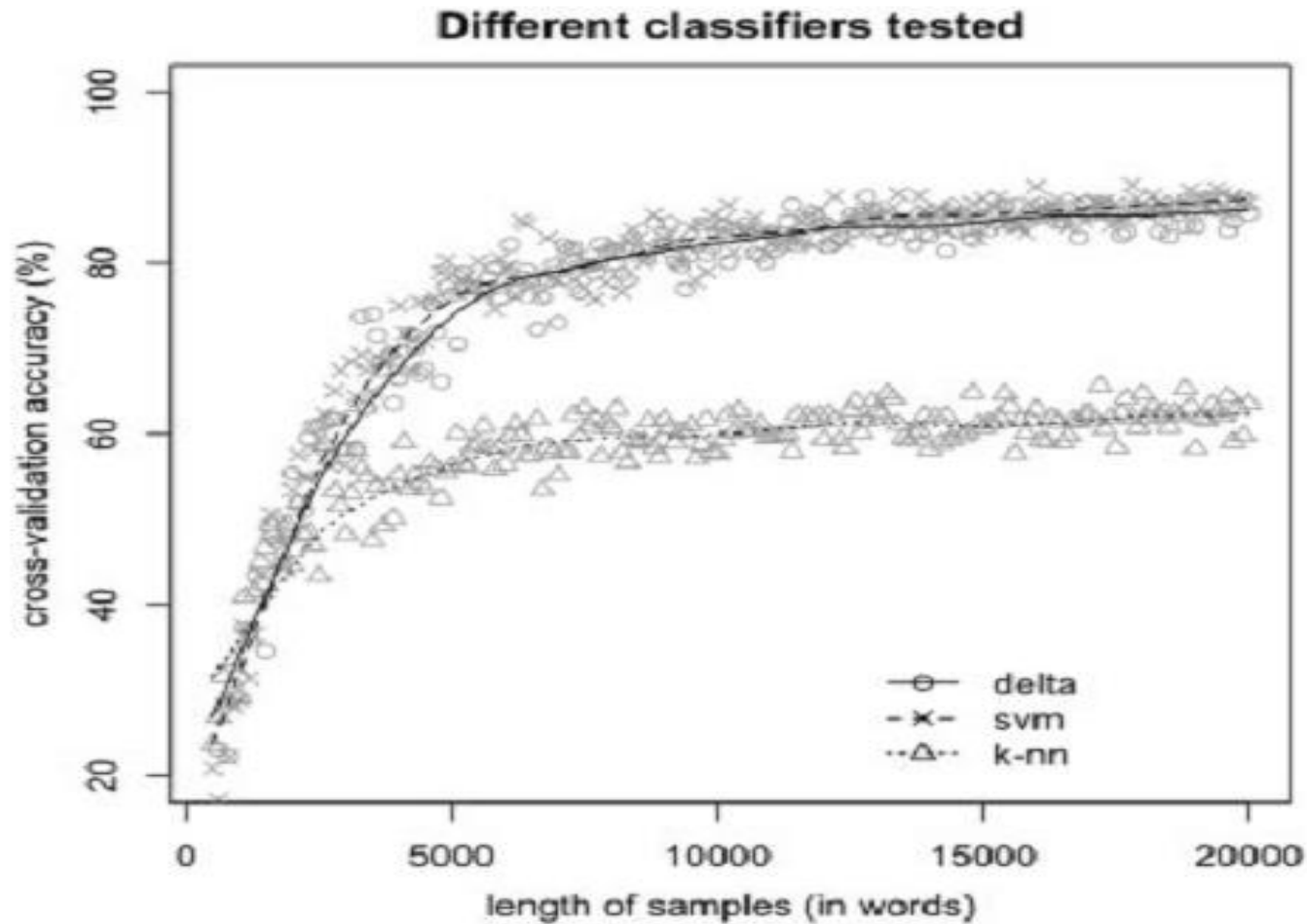


[Giovanni Morelli](#), Public domain, via Wikimedia Commons

1. the
2. and
3. of
4. to
5. a
6. i
7. in
8. he
9. was
10. it
11. that
12. you
13. his
14. her

“It has been noted that the switch from content words to function words in authorship attribution studies has **an interesting historic parallel in art-historic research.** [...] Giovanni Morelli (1816-1891) was among the first to suggest that the attribution of, for instance, a Quattrocento painting to some Italian master, could not happen based on ‘content’ [...] Morelli thought it better **to restrict an authorship analysis to discrete details such as ears, hands and feet**” (Kestemont 2014)

CORPUS SELECTION (TEXT DIMENSIONS)

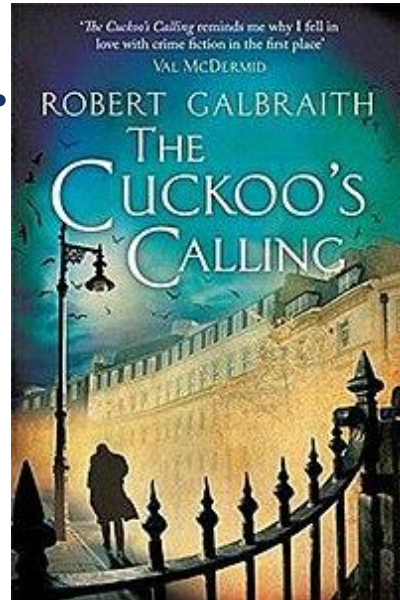


Minumun text length for a reliable stylometric analysis is about 5,000 words (Eder 2015)

APPLICATIONS IN LITERARY STUDIES

J.K.
Rowling

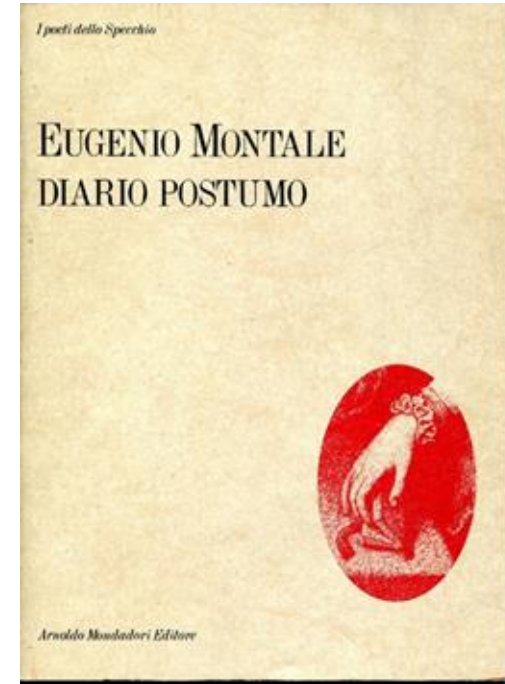
(Juola, 2015)



(Tuzzi and Cortelazzo, 2018)



(Italia and Canettieri, 2013)



APPLICATIONS IN LITERARY STUDIES

ON LATE STYLE

MUSIC AND LITERATURE
AGAINST THE GRAIN

EDWARD W. SAID

"These studies ... buzz with excitement and intelligence and demonstrate what his admirers already knew, the extraordinary range of Said's intellectual interests."
—Frank Kermode, *London Review of Books*



Does Late
Style Exist?

APPLICATIONS IN LITERARY STUDIES

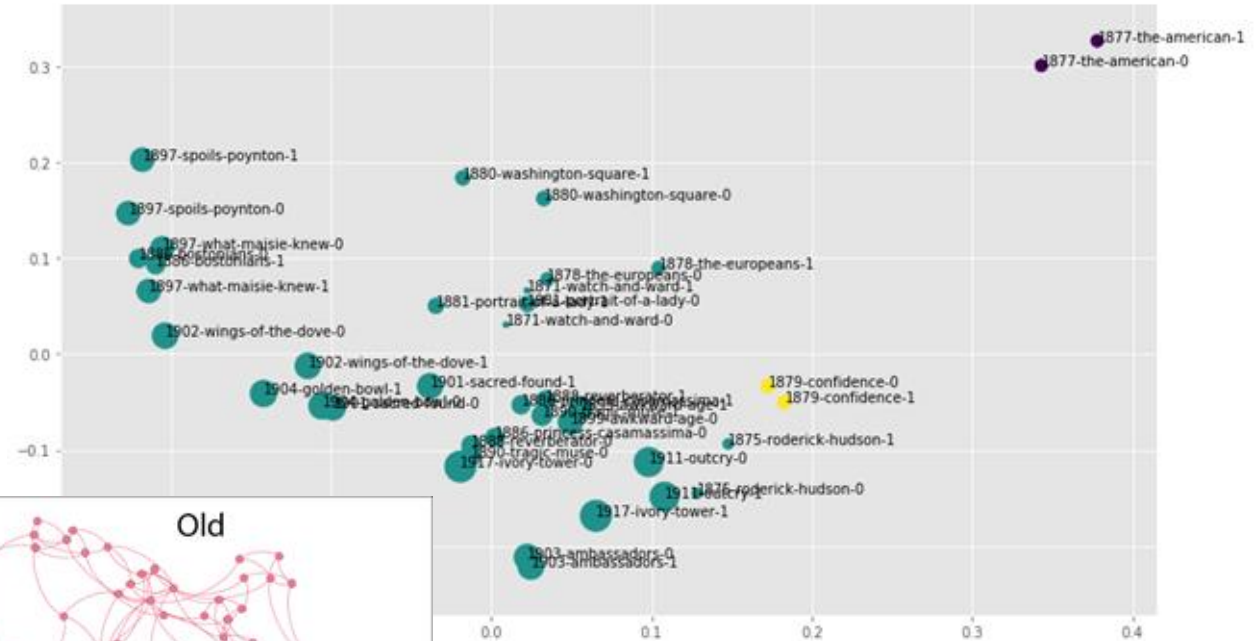
ON LATE STYLE

MUSIC AND LITERATURE
AGAINST THE GRAIN

EDWARD W. SAID

Does Late
Style Exist?

(Reeve,
2018)



(Rebora and
Salgaro, 2018)

Young

Kafka's early works

Middle

Old

HANDS-ON!!!

Stylo

(Eder et al. 2016)

README.md readme update 2 years ago

README.md

stylo: R package for stylometric analyses

Authors: Maciej Eder*, Mike Kestemont, Jan Rybicki, Steffen Pielström
License: [GPL-3](#)

CRAN 0.7.4 build passing downloads 109K downloads 1182/month

This package provides a number of functions, supplemented by a GUI, to perform various analyses in the field of computational stylistics, authorship attribution, etc.

Refer to the [Computational Stylistics Group webpage](#), especially the subpage [Projects](#), to get some ideas about possible applications of the package `stylo`.

Citation

If you find the package `stylo` useful and plan to publish your results, please consider citing the following paper:

Eder, M., Rybicki, J. and Kestemont, M. (2016). Stylometry with R: a package for computational text analysis. *R Journal*, 8(1): 107-21. <https://journal.r-project.org/archive/2016/RJ-2016-007/index.html>

Installation

There are four ways of installing `stylo`:

1. from CRAN repository
2. from the GitHub repository, via the package `devtools`
3. from a locally downloaded file
4. building the package directly from source files

Contributors 12



Languages

R 100.0%

Compiled in the
R programming language

<https://github.com/computationalstylistics/stylo>

THE “R” PROGRAMMING LANGUAGE

- A programming language for statistics and graphics
- A free software (licence GNU GPL)
- Created in 1995 by
 - Ross Ihaka (New Zealand)
 - and Robert Gentleman (Canada)
- The name? See the initials of the creators

```
rsimone@rsimone-Inspiron-13-5378:~$ R
R version 3.4.4 (2018-03-15) -- "Someone to Lean On"
Copyright (C) 2018 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

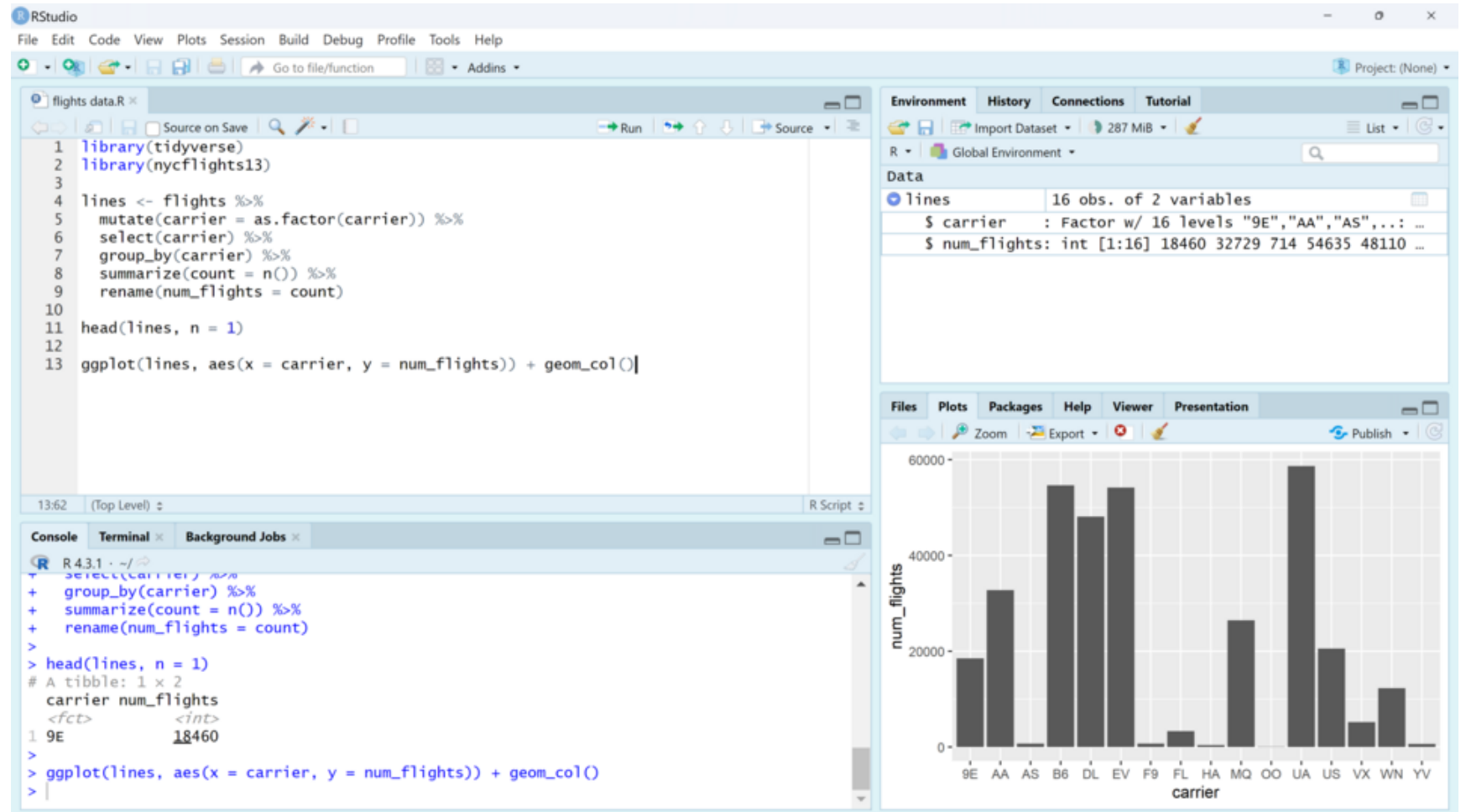
R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> a <- 1:10
> a
[1] 1 2 3 4 5 6 7 8 9 10
> for(i in 1:10)
+ print(i:10)
[1] 1 2 3 4 5 6 7 8 9 10
[1] 2 3 4 5 6 7 8 9 10
[1] 3 4 5 6 7 8 9 10
[1] 4 5 6 7 8 9 10
[1] 5 6 7 8 9 10
[1] 6 7 8 9 10
[1] 7 8 9 10
[1] 8 9 10
[1] 9 10
[1] 10
>
```

POSIT

A graphical
interface
(desktop or
cloud) to use
the R
programming
language



Instructions

1. Connect to posit.cloud
2. Sign up (Cloud Free Plan; Sign Up with Google)
3. Create a new project from Git Repository
4. Copy-paste the clone link of the EnExDi2024 GitHub repository
5. Click on OK!

